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Probing millikelvin temperature sensitivity in chiral nanoparticles via optical forces

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Note: This paper is part of the Special Topic on Photothermal Materials.

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ABSTRACT

With increasing interest in utilizing nanostructures as nanoscale heat sources, the ability to precisely measure photothermal effects at the nanoscale has become increasingly significant. Techniques based on fluorescence or Raman signals often suffer from challenges in accurate calibration; far-field imaging methods are limited by diffraction-limited spatial resolution; and electron microscopy requires vacuum conditions, restricting *in situ* applicability. In contrast, tip-based measurement techniques offer sub-diffraction spatial resolution under ambient conditions, making them well-suited for nanoscale photothermal mapping. In this study, we employ tip-based optical force nanoscopy combined with phase-informed decomposition to investigate the origin of the photothermal force, enable nanoscale mapping, and evaluate temperature sensitivity. Our system achieves a temperature sensitivity of ~ 0.1 K without necessitating an additional temperature-sensitive layer. We anticipate that our approach has the potential to serve as a versatile platform for investigating localized thermal effects in fields such as semiconductors, nanophotonics, and photocatalysis.

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INTRODUCTION

Temperature is a physical quantity that describes the motion of atoms or molecules within a material and indicates its internal energy. However, accurate temperature measurement remains a significant challenge due to the rapid dissipation of thermal energy and the difficulty in maintaining thermal equilibrium. Nanophotonics has recently emerged as a versatile platform for sensing,^{1,2} imaging,³ photocatalysis,⁴ energy harvesting,⁵ and biomedical applications.^{6,7} Beyond its ability to concentrate and enhance light-matter interactions, nanophotonics also enables the conversion of optical energy into various forms. Traditionally, the photothermal conversion of light into heat has been considered a parasitic loss. However,

the growing interest in employing nanoparticles as localized heat sources^{8,9} has highlighted the importance of nanoscale temperature visualization, particularly in emerging applications, such as photoacoustic imaging,¹⁰ photothermal catalysis,^{11,12} and photothermal therapy.¹³

To achieve microscopic temperature mapping, various fluorescence-based thermometry techniques have been developed, which rely on temperature-dependent changes in fluorescence intensity,¹⁴ lifetime,¹⁵ and anisotropy.¹⁶ Despite their potential, these methods face inherent limitations. Fluorescence has a low dynamic range in temperature due to the loss of signal characteristics at high temperatures and is also affected by various extrinsic factors, including pH, molecular concentration, surface chemistry,

and energy transfer processes,¹⁷ complicating accurate temperature calibration. Raman thermometry faces comparable constraints.¹⁸ Alternative approaches include quantitative phase microscopy,¹⁹ which visualizes temperature through temperature-induced changes in the refractive index, and photothermal microscopy,²⁰ which employs thermal lensing effects along with pump-probe measurements. While these methods enable cellular-scale thermal imaging and, in the case of photothermal microscopy, can be combined with infrared spectroscopy for chemical contrast, they are inherently limited by diffraction due to their far-field imaging nature. Electron energy loss spectroscopy overcomes the diffraction limit and enables temperature mapping with nanometer-scale spatial resolution;²¹ however, its applicability is constrained by the need for electron microscopy under high-vacuum conditions.

In contrast, tip-based thermometry techniques offer superior spatial resolution determined by the tip size, allowing access to sub-diffraction-limited thermal features. Atomic force microscopy-infrared (AFM-IR) and photoinduced force microscopy (PiFM), which combine atomic force microscopy with infrared spectroscopy, enable simultaneous acquisition of thermal expansion and chemical information at the nanoscale.^{22–26} Apertureless near-field scanning optical microscopy revealed that photothermal expansion of PMMA was the predominant effect, surpassing the contribution of optical gradient forces in the infrared region.²⁷ In addition, aperture-type near-field scanning optical microscopy allows the direct observation of the nonlinear features in light absorption and near-field enhancement.²⁸ Recently, decoupled optical force nanoscopy has been introduced, employing phase-informed decomposition to distinguish the optical gradient and photothermal force.²⁹ This technique has demonstrated the potential for investigating ultrafast photothermal response with high spatial resolution.

Here, we present a methodology for temperature quantification based on the correlation between photothermal force and thermally induced expansion, as measured by optical force nanoscopy. We systematically investigate the underlying origin of the photothermal force and derive the theoretical temperature sensitivity limit of

the technique. Leveraging this framework, we further demonstrate polarization-dependent thermal responses in chiral gold nanoparticles under circularly polarized light, highlighting the capability of the approach for nanoscale thermometric analysis with sub-diffraction spatial resolution.

RESULTS AND DISCUSSION

Thermal expansion and photothermal force

It might seem unusual that the thermal expansion of a metal film just a few tens of nanometers thick can result in a measurable displacement of several tens of nanometers. However, as the temperature of the glass substrate onto which the metal film is deposited increases, the thermal expansion of the glass substrate itself can also reach several tens of nanometers. The typical thickness of a glass substrate is around 150 μm . The thermal expansion coefficient of a BK7 glass substrate is $7.1 \times 10^{-6}/\text{K}$, meaning that a uniform increase of 1 $^{\circ}\text{C}$ in temperature leads to an expansion of ~ 1 nm.

Figure 1(a) shows a schematic illustration of the photothermal expansion and the resulting photothermal force of a gold thin film deposited on a glass substrate. When light is absorbed by the gold film, it generates heat, leading to the thermal expansion of both the gold film and the glass substrate. The AFM cantilever maintains a constant distance between the tip and the sample, where the deflection signal reflects the photothermal force. Figure 1(b) illustrates the temperature profile along the z -axis under uniform illumination, which is obtained from the following equation³⁰ in cylindrical coordinates:

$$T(z) = \frac{q_0}{4\pi\kappa} \int_0^R \int_0^{2\pi} \frac{1}{\rho^2 + z^2} \rho d\theta d\rho = \frac{q_0}{2\kappa} \left(\sqrt{R^2 + z^2} - z \right), \quad (1)$$

where q_0 and κ represent the heat source and thermal conductivity, respectively, and R is the radius of the illuminated area. The resulting thermal expansion can be calculated using the following equation:²⁸

$$L = L_{\text{gold}} \left(1 + 3\alpha_{\text{gold}} \Delta T(z) \right) + \int_{-L_{\text{gold}} - L_{\text{BK7}}}^{-L_{\text{gold}}} \left(1 + 3\alpha_{\text{glass}} (T_{\text{max}} - T_{\text{min}}) \left(\sqrt{1 + \left(\frac{z}{R} \right)^2} - \frac{|z|}{R} \right) \right) dz, \quad (2)$$

where L_{gold} and L_{BK7} represent the thickness of the gold film and the glass substrate, respectively, and α denotes the thermal expansion coefficient of the materials. The thermal expansion coefficients used in the analysis are $14 \times 10^{-6} \text{ 1/K}$ for gold and $7.1 \times 10^{-6} \text{ 1/K}$ for glass. Since the illuminated area covers only a small portion of the total sample surface, the volumetric expansion is approximated as linear expansion.

To verify the thermally induced axial shift of the AFM cantilever, we positioned the cantilever onto the sample surface and cycled the laser on and off at intervals of 1.5 s. The experiments were performed on a 160 nm gold film with zero transmittance to prevent the heating (excitation) beam from being detected by the AFM detector. Under this condition, the heating beam is unable to induce

light-matter interaction at the tip-sample junction. Therefore, we attribute the observed axial shift solely to the thermal expansion. Circularly polarized light at a wavelength of 637 nm was used as the heating beam. As illustrated in Fig. 1(c), the z sensor, which indicates the tip-sample distance, maintains a constant value, while the tip dwells on the surface (green-shaded area). Meanwhile, the amplitude signal, reflecting the relative z position of the cantilever, fluctuates up and down depending on whether the laser light is on or off. Figure 1(d) shows the axial shift of a 160 nm thick gold film on a glass substrate under varying incident powers. The amplitude-to-nanometer conversion coefficient (Amp InvOLS) was measured to be 306.89 nm/V. The measured axial shift increased with increasing power. The lowest laser power, denoted as $1\times$, corresponds to an

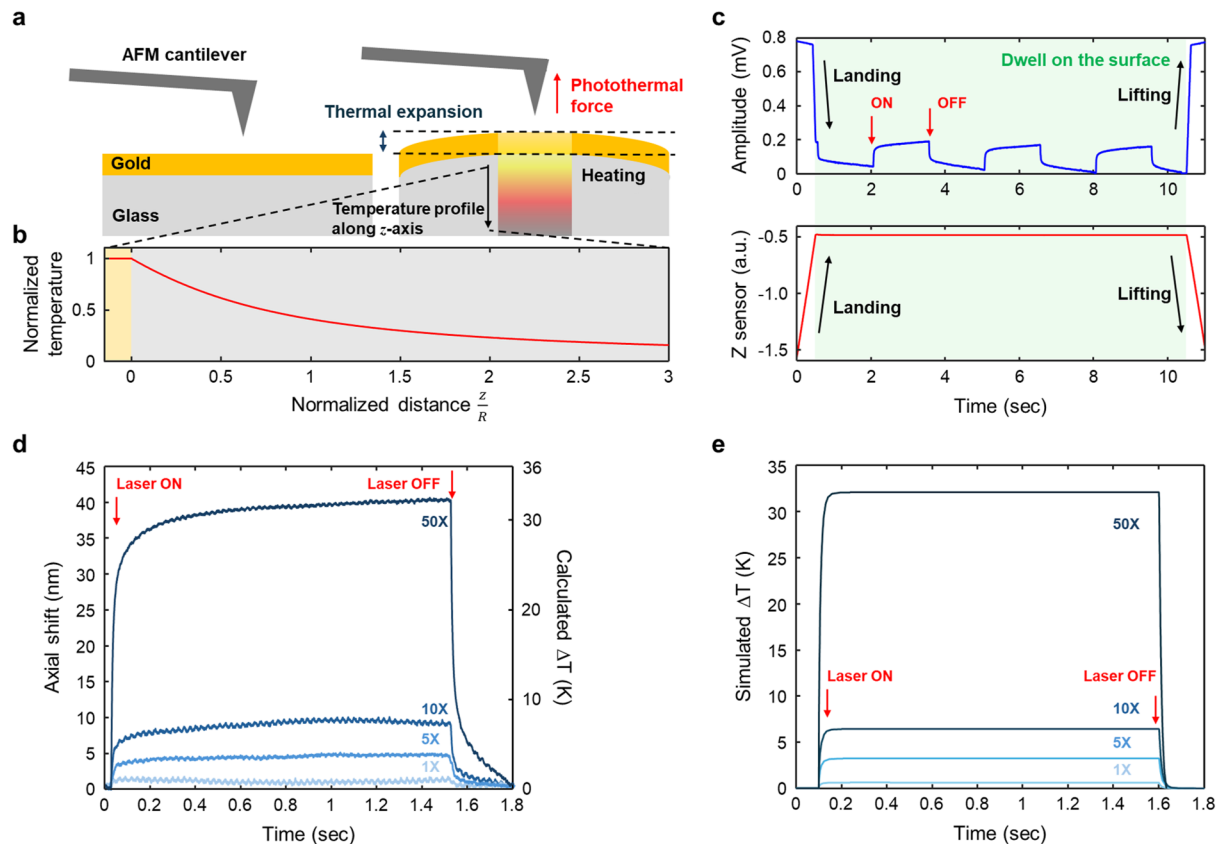


FIG. 1. Origin of photothermal force. (a) Schematic illustration of the photothermal force, showing thermal expansion upon optical heating of a gold film and the underlying glass substrate. Heat dissipates through the substrate, and a scanning probe measures the overall thermal expansion of the film. (b) Analytical solution of the temperature profile along the z-axis under uniform illumination. R denotes the radius of the illuminated field of view. (c) Amplitude of the cantilever deflection and z sensor signal as functions of time during a cycle, in which the probe engages, dwells, and withdraws. During the dwell phase, the laser is toggled on and off to allow measurement of the photothermal force. (d) Measured temporal response of the axial shift of the gold film under various incident laser powers. “1 $\times$ ” corresponds to an incident intensity of $1.5 \times 10^5 \text{ W/m}^2$. The numbers indicate the power ratios relative to this baseline. (e) Simulated temporal response of the gold film under varying incident laser powers.

incident intensity of $1.5 \times 10^5 \text{ W/m}^2$. At 1 $\times$ power, the average axial shift was 1.19 nm with a standard deviation of 0.223 nm. Increasing the power to 5 $\times$ resulted in a mean shift of $4.65 \pm 0.228 \text{ nm}$, while at 10 $\times$ power, the shift further increased to $10.38 \pm 0.547 \text{ nm}$. At the highest power level, 50 $\times$, the mean axial shift reached $39.26 \pm 0.921 \text{ nm}$. In reported values, the number preceding the $\pm$ symbol represents the mean, while the number following it denotes the standard deviation. The standard deviation relative to the mean stays within approximately 2%–5% for power levels above 5 $\times$. The axial shift of 39.26 nm corresponds to the temperature increment of 32 K, which is in excellent agreement with the simulation results shown in Fig. 1(e). Our system enables the detection of an axial shift as small as 1.19 nm at the lowest power, corresponding to a temperature elevation of 0.9 K. The temperature detection limit of the optical force nanoscopy system is closely related to the axial resolution of the AFM, which is $\sim 0.05 \text{ nm}$. Taking into account the thermal expansion of the glass substrate and the gold thin film, this resolution corresponds to a temperature change of $\sim 0.04 \text{ K}$. While this value represents the theoretical lower limit of the system’s temperature

sensitivity, such minute temperature variations are highly susceptible to environmental fluctuations and thermal noise. Therefore, these measurements should be interpreted with caution.

Characterization of chiral nanoparticles

To determine the temperature sensitivity of a single particle, we employed chiral gold nanoparticles in conjunction with circularly polarized light. The handedness of the circular polarization affects the absorption properties of the chiral gold nanoparticles, resulting in varying temperature increases. By measuring these temperature differences, we further evaluated the temperature sensitivity of the optical force nanoscopy system. In addition, a key advantage of using circularly polarized light is its ability to generate a consistent optical response regardless of the orientation of the gold nanoparticles.

The chiral gold nanoparticles were synthesized using a seed-mediated growth method using rhombicuboctahedron nanoparticles.^{31,32} A scanning electron microscopy (SEM) image of the chiral gold nanoparticles is presented in Fig. 2(a). The

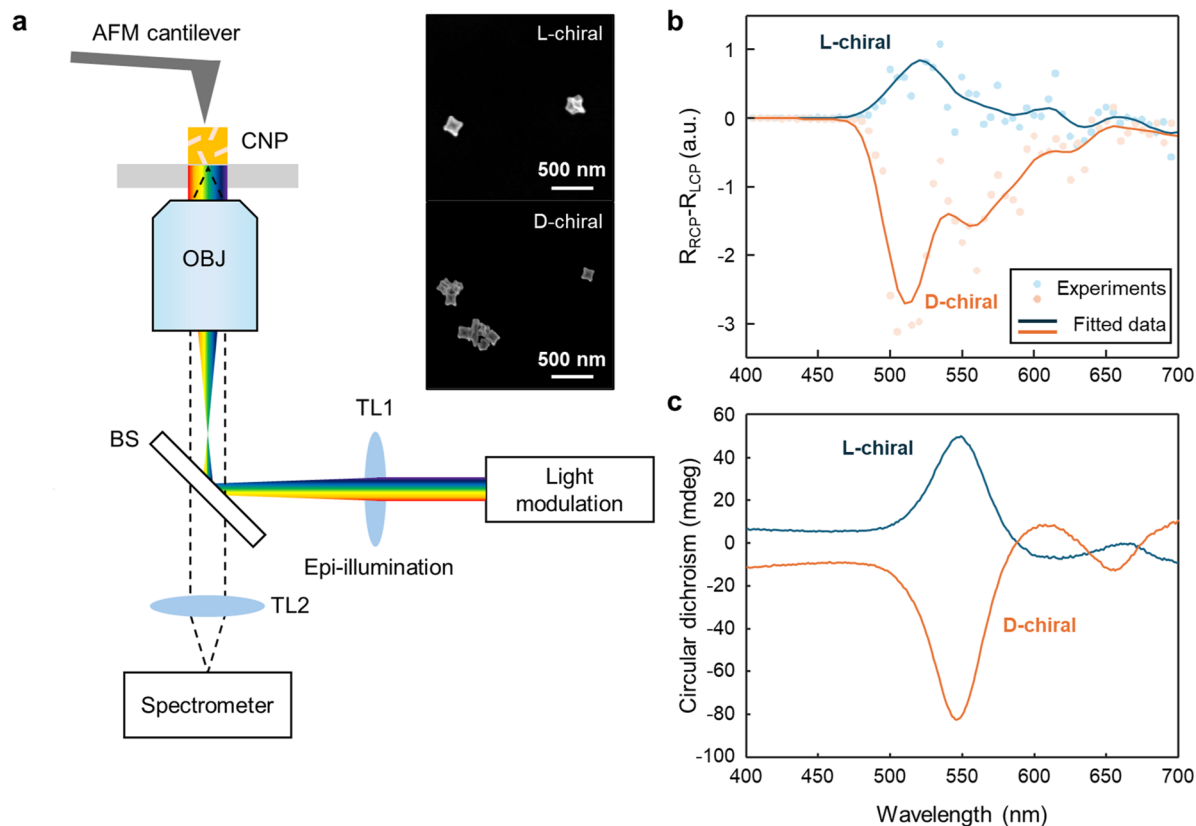


FIG. 2. Optical force nanoscopy setup and characterization of chiral gold nanoparticles. (a) Schematic illustration of the optical setup and SEM images of the chiral gold nanoparticles (CNP: Chiral nanoparticles, OBJ: Objective lens, BS: Beam splitter, TL: Tube lens). (b) Reflectance difference spectra of the chiral gold nanoparticles measured on a glass substrate. (c) Circular dichroism spectra of the chiral gold nanoparticles in aqueous solution.

nanoparticle concentration is 5.6×10^9 particles/ml. The particles were diluted to 10% concentration using deionized water and drop-cast onto a glass substrate, followed by solvent evaporation under vacuum. Residuals were then removed by rinsing with deionized water.

Figure 2(a) also shows the optical measurement setup used to characterize the optical properties of the chiral nanoparticles. A supercontinuum laser was used to generate a white light source centered at 550 nm with a spectral bandwidth of 300 nm. The beam passes through a polarization modulation module consisting of a linear polarizer, a half-wave plate, and a quarter-wave plate before reaching the sample. The incident light is wide-field illumination with normal incidence. Upon interacting with chiral nanoparticles, the reflected light is collected and analyzed using a spectrometer. Spectra were recorded ten times with an integration time of 100 ms at 2-nm intervals using ten sampling boxes, and the average value was subsequently calculated. To eliminate the influence of the substrate, the spectrum of a bare glass substrate without nanoparticles was also measured and used as a background for subtraction. The measured data were smoothed using a moving average filter.

The differential reflectance spectra between left- and right-handed circularly polarized light are presented in Fig. 2(b). The

L-chiral nanoparticles exhibit an overall positive differential reflectance with a peak at 520 nm, while the D-chiral counterparts show a negative differential reflectance with a dip at 510 nm. These spectral features are blueshifted compared to the liquid-state circular dichroism spectra obtained using a commercial spectrometer (JASCO J-1500), as shown in Fig. 2(c). This discrepancy arises from the differences in the surrounding environment: the measurements in Fig. 2(b) were conducted at the glass–air interface, whereas those in Fig. 2(c) were performed in aqueous solution using a commercial circular dichroism spectrometer. There is no measurable difference in circular dichroism between L-chiral and D-chiral gold nanoparticles at a wavelength of 640 nm [see Fig. 2(c)]. However, we observed a small difference in differential reflectance at the same wavelength, as shown in Fig. 2(b). Since such small signal differences are useful for evaluating sensitivity, we chose a wavelength of 637 nm to induce photothermal force.

Photothermal force of a chiral gold nanoparticle

In the optical force nanoscopy system, the incident light was modulated using a function generator at a frequency f_0 , which is identical to the AFM cantilever's resonant frequency. The AFM

cantilever was driven at a frequency of 100 Hz above its resonant frequency to ensure operation in the attractive regime. This modulation enables the decomposition of the total optical force into three components: optical gradient force, photothermal force, and background force—typically attributed to photoacoustic effects—through phase-informed analysis.²⁹ Figure 3 presents the measured photothermal force from an L-chiral gold nanoparticle. Figure 3(a) shows the AFM topography of a single L-chiral gold nanoparticle. The arrows indicate the vertices of the chiral nanoparticle, which are also identifiable in the photothermal force map discussed later. The maximum measured height reaches up to 225 nm. Figure 3(b) presents a simulated temperature distribution of a 200 nm chiral cube that mimics the geometry of an L-chiral gold nanoparticle under right-handed circularly polarized (RCP) light. The incident optical intensity was set to $\sim 8 \times 10^6 \text{ W/m}^2$, matching the experimental conditions. The thermal conductivities used in the simulation were 0.025 W/m K for air, 317.98 W/m K for gold, and 1.114 W/m-K for the BK7 glass substrate. A temperature rise of 1.25 K was observed under RCP illumination, whereas a comparatively lower increase of 0.97 K was observed under left-handed circularly polarized (LCP) light.

Figures 3(c) and 3(d) show the photothermal force distribution of an L-chiral gold nanoparticle under RCP and LCP illumination, respectively. The maximum photothermal force was measured to be 142 pN for RCP, whereas 117 pN for LCP. The four vertices of the chiral nanoparticle, indicated by arrows in Fig. 3(a), are clearly identifiable in the photothermal force maps. Notably, the recessed areas of the chiral nanoparticle exhibit significantly lower photothermal force compared to the protruding features. In addition,

consistent with the simulation results, the photothermal force within the nanoparticle is substantially higher than in the surrounding regions. Figure 3(e) presents the topographic profiles along the x - and y -axes of the chiral nanoparticle. Along the x -axis, two protruding features are observed, while three distinct protrusions are noted along the y -axis. Although these structural features are not fully replicated, the simulations depicted in Figs. 3(f)–3(h) clearly reveal the corresponding photothermal force distributions—two protrusions along the x -axis and three along the y -axis. These patterns arise from variations in temperature distribution caused by air gaps within the nanoparticle structure, which are dependent on the shape of the particle. The profiles shown in Figs. 3(g) and 3(h) were smoothed using a moving average filter. The average photothermal force over the field of view, excluding the nanoparticle region, was measured to be 13.3 pN under RCP and 15 pN under LCP. However, when focusing exclusively on the interior region of the chiral nanoparticle, the photothermal force significantly increases to 63.9 pN for RCP and 53.9 pN for LCP, indicating a pronounced polarization-dependent response.

This result suggests that heat dissipation is ultimately governed by the surrounding environment, and in the absence of a pronounced temperature gradient, the overall region experiences nearly uniform photothermal force and thermal expansion. The simulated temperature difference of 0.28 K corresponds to a photothermal force difference of $\sim 11.7 \text{ pN}$, which was obtained from the difference in the photothermal force between the RCP and LCP. A photothermal force of 11.7 pN results in an observed axial shift of 0.37 nm, as determined by the back-calculation with a spring constant of

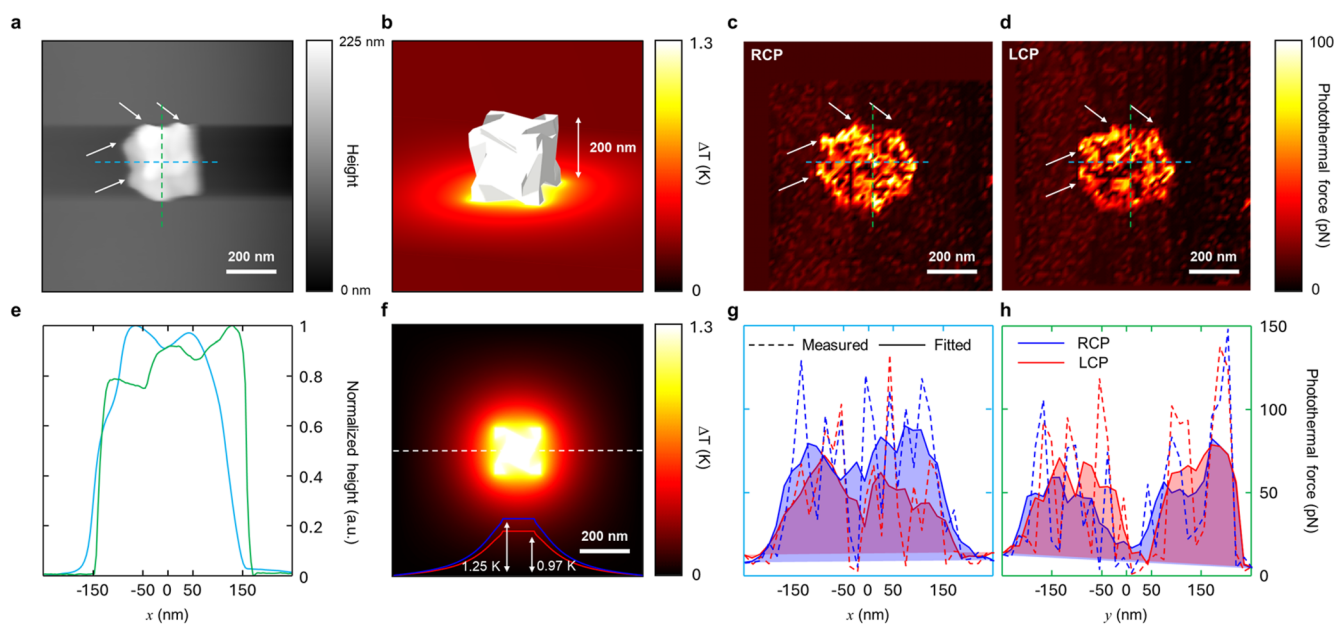


FIG. 3. Photothermal force mapping of an L-chiral gold nanoparticle. (a) AFM topography of an L-chiral gold nanoparticle. (b) Simulated temperature distribution of the L-chiral gold nanoparticle under right-handed circularly polarized (RCP) light. [(c) and (d)] Measured photothermal force maps of L-chiral gold nanoparticle under RCP and left-handed circularly polarized (LCP) lights, respectively. (e) Height profile of the L-chiral gold nanoparticle (x -axis: sky blue, y -axis: green). (f) Simulated temperature distribution at $z = 200 \text{ nm}$ and corresponding temperature profile along the x -axis (RCP: blue, LCP: red). [(g) and (h)] Measured photothermal force profiles along the x - and y -axes. The scale bar represents 200 nm.

2.5 nN/nm and frequency response of the cantilever. The resonant frequency of cantilever f_0 was 67.244 kHz, and the drive frequency of the cantilever f_d was 67.344 kHz. In the analysis, the stiffening and softening of the cantilever due to the illumination were ignored. An axial shift of 0.37 nm corresponds to the thermal expansion of both the gold nanoparticle and the glass substrate, resulting from a 0.3 K temperature increase, consistent with the earlier analysis. This close agreement between the simulation and experiment confirms the consistency and validity of the results.

Importantly, the measured displacement surpasses the axial resolution of the AFM used in the system. With an AFM axial resolution of 0.05 nm, which corresponds to a temperature increment of 0.04 K based on the thermal expansion calculations, we estimate that the minimum detectable optical force of the system is ~ 1.6 pN. However, we remain cautious about directly correlating absolute photothermal force values with precise temperature estimates. Although the temperature distribution within the gold nanoparticle is expected to remain uniform due to the high thermal conductivity of gold, the measured photothermal force also includes contributions from mechanical noise. This additional signal makes it challenging to assign a precise temperature to a given force value with high confidence. Therefore, determining temperature solely from individual photothermal force measurements may lead to inaccurate results. Instead, extracting temperature by comparing measured photothermal forces with calibrated reference

data, as presented in this study, offers a more reliable and robust approach.

Temperature sensitivity of the optical force nanoscopy

Figure 4 presents the average photothermal force and corresponding photothermal force maps within the chiral nanoparticle and across the entire field of view as functions of incident light power and polarization. The lowest intensity, used as a reference for calculating the power ratio, was 1.5×10^5 W/m². As previously discussed, Fig. 4(a) confirms that the photothermal force within the L-chiral nanoparticle is higher under RCP. However, when the power ratio is below 30, it becomes difficult to distinguish the photothermal force originating from the nanoparticle from the background signal. This is due to an insufficient temperature rise, resulting in limited contrast in the photothermal force distribution. For example, at a power ratio of 20, the average photothermal force inside the nanoparticle is 13.3 pN for RCP and 17 pN for LCP, while the background levels are 11.1 pN for RCP and 11.2 pN for LCP. This leads to a modest difference of only 2.2 pN compared to the background level, which is indistinguishable from the noise. In contrast, at a power ratio of 30, the photothermal force within the nanoparticle rises to 36.2 pN for RCP and 25.8 pN for LCP, with the background remaining at 13.6 pN for RCP and 13.8 pN for LCP. This results in a

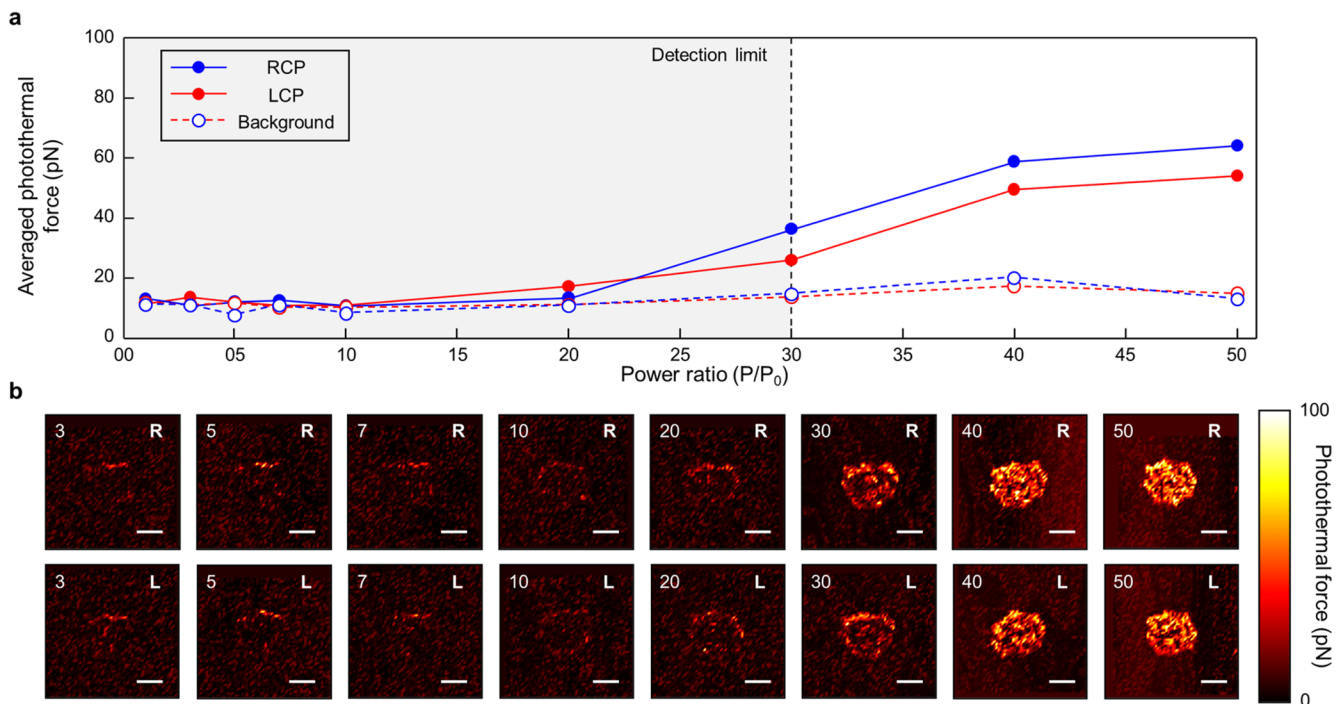


FIG. 4. Temperature sensitivity mapped on chiral gold nanoparticles. (a) Average photothermal forces measured in the chiral gold nanoparticle region and the surrounding background as a function of incident optical power. The background signal was obtained from regions excluding the nanoparticle. To exclude areas of low photothermal force caused by internal air gaps within the nanoparticle, we calculated the average photothermal forces in two steps: first, a preliminary average was calculated over the nanoparticle region; then, a final average was calculated with the values that exceeded the preliminary average to exclude the air gaps. (b) Photothermal force maps under varying incident powers and circular polarizations. The top-left number in each panel indicates the power ratio, and the top-right R/L indicates the handedness of the circular polarization. The scale bar represents 200 nm.

clear and reliable photothermal force difference of 10.6 pN between RCP and LCP, respectively. This observation is further supported by the photothermal force maps illustrated in Fig. 4(b). At power ratios below 20, while the general shape of the nanoparticle is somewhat visible, it is more reasonable to attribute this to edge artifacts typically encountered in tip-based measurements rather than true photothermal force signals. In contrast, at power ratios of 30 and above, not only is the nanoparticle morphology clearly defined, but a uniformly high contrast photothermal force distribution is also observed within the particle.

Based on these findings, we define the temperature detection limit of the optical force nanoscopy system as the temperature difference between the RCP and LCP, which is ~ 0.17 K. This value was obtained from the simulation results using a power ratio of 30. From the measurements, the photothermal force difference of 10.6 pN corresponds to an axial shift of 0.33 nm. Based on thermal expansion calculation, this shift corresponds to a temperature increment of 0.26 K, corroborating the simulation results. Our analysis, illustrated in Figs. 1, 3, and 4, demonstrates that the temperature detection limit of the optical force nanoscopy system is close to 0.1 K. However, it is important to note that this value may vary due to the inherent limitations of all tip-based measurements, in particular, the dependence of data acquisition on the specific AFM tip used.

CONCLUSION

In this study, we show that the origin of the photothermal force measured by optical force nanoscopy arises from the axial shift of the cantilever, which is primarily driven by the thermal expansion of both the nanoparticle and the substrate. The axial shift, obtained from experimentally measured photothermal force, can be converted into temperature through thermal expansion analysis. This inferred temperature shows excellent agreement with the predicted values obtained from photothermal simulations. In addition, we establish the temperature detection limit of the system based on the axial resolution of the AFM. The spatial resolution of our method is solely determined by the size of the tip, which readily surpasses the optical diffraction limit.

While our findings show that nanoscopic temperature mapping with a sensitivity on the order of 100 mK is achievable, several challenges remain to be addressed. Although optical force nanoscopy operates in the attractive regime, it remains sensitive to sample geometry, as evidenced by the edge artifacts observed in Fig. 4(b). Another limitation is the requirement for a reference point when measuring forces resulting from thermal expansion. If the entire field of view experiences a uniform temperature increase, axial shifts may occur, but without a relative baseline, these shifts would be indistinguishable. This constraint is not exclusive to our method; it is a common limitation among many indirect temperature detection techniques, most of which rely on comparing the “on” and “off” states to establish a reference.

Our analysis sheds new light on the tip-based thermal response measurements. In conventional nanoscale thermometric techniques, the contribution of substrate thermal expansion is often disregarded.³³ However, by employing phase-informed decomposition to selectively isolate the photothermal force, it becomes feasible to achieve temperature sensitivity on the order of 0.1 K without the need for an additional temperature-sensitive layer. This approach

offers a distinct advantage by enabling high-resolution thermal measurements under ambient, non-vacuum conditions. Consequently, it holds significant promise for broad application in emerging fields, where localized thermal effects at the nanoscale are critically important, such as semiconductor and nanophotonic device engineering and photocatalysis.

MATERIALS AND METHODS

Optical force measurements

The decoupled optical force nanoscopy system was implemented using an inverted optical microscope (IX81, Olympus) integrated with an atomic force microscope (MFP-3D-BIO, Oxford Instruments Asylum Research). A 637 nm diode laser (OBIS 637 nm, 140 mW, Coherent) was employed for measurements involving metallic thin films and photothermal force detection. The diode laser output was modulated in power, frequency, and duty cycle using a function generator (33250A 80 MHz Function/Arbitrary Waveform Generator, Agilent). The polarization state of the laser was controlled by the combination of a half-wave plate (AHWP10M-600, Thorlabs) and a quarter-wave plate (AQWP10M-580, Thorlabs). The beam was then focused onto the back focal plane of the objective lens (UAPON100XOTIRF, Olympus) via a tube lens, enabling wide-field illumination of the sample surface. An AC240 cantilever (4XC-NN, OPUS) was used for the AFM measurements. The deflection signal of the AFM cantilever was acquired using a lock-in amplifier (Signal Recovery 7280 DSP Lock-in Amplifier). The lock-in signal was acquired at the sampling frequency of 200 Hz, and the scan rate of the AFM was 20 Hz. An averaging window of ten data points was established to quantitatively assess the magnitude and phase of the AFM deflection signal at each pixel. Furthermore, averaging the trace and retrace results effectively eliminates the ascending and descending artifacts that arise when the AFM tip is positioned on the structure.

Photothermal simulation

Photothermal simulations were carried out using the RF and heat transfer modules in COMSOL Multiphysics. For gold thin-film structures, periodic boundary conditions were applied, with Floquet boundary conditions for the wave equation and zero heat flux boundary conditions for the heat transfer equation. For a gold nanoparticle, a two-step wave equation followed by a one-step heat transfer calculation was employed to reduce computational cost. In this approach, a plane wave was first applied to the periodic structure, and the resulting field was then used as the incident wave for a single gold nanoparticle, where the periodic boundary conditions were replaced with perfectly matched layers to emulate an open boundary environment. The subsequent heat transfer calculation was performed using this configuration. Since plasmonic structures do not induce extremely high temperatures under moderate illumination conditions, interfacial thermal resistance was not considered in our simulations.

Chiral nanoparticle synthesis

The Au chiral nanoparticles were synthesized following a previously reported seed-mediated growth method using rhombicuboctahedron (RCO) nanoparticles.³¹

Synthesis of 2 nm gold seeds

In a typical synthesis, 32 mg of CTAC was dissolved in 975 μl of H_2O , followed by the addition of 25 μl of 10 mM HAuCl_4 solution. To this solution, 45 μl of freshly prepared ice-cold 20 mM NaBH_4 solution was rapidly injected under stirring, which was then kept still under 30 °C for 1 h.

Synthesis of Au RCO nanoparticles

In a typical synthesis, two identical growth solutions (Vial A and Vial B) were prepared, each containing 320 mg of CTAC, 9.472 ml of H_2O , 250 μl of 10 mM HAuCl_4 , and 30 μl of 1 mM KI. After mixing, 220 μl of 40 mM AA solution was added to each vial. After this, 55 μl of previously prepared 2 nm gold seed was added into Vial A under stirring. The solution changed to pink color within 10–15 s, and 55 μl of this mixture was then immediately injected into Vial B under stirring. Vial B was left undisturbed at 30 °C for 15 min to allow the growth of RCO nanoparticles. The RCO nanoparticles were cleaned twice using H_2O by spinning at $6700 \times g$ for 2.5 min and dispersed in 1 ml of H_2O .

Synthesis of Au chiral nanoparticles

In a typical synthesis, 0.8 ml of 0.1M CTAB and 100 μl of 10 mM HAuCl_4 were combined in a vial containing 3.6 ml of H_2O . To this vial, 475 μl of 0.1M AA, 6 μl of 0.1 mM of L- or D-cysteine, and 3 μl of 1 mM $\text{Cu}(\text{NO}_3)_2$ were added sequentially. The mixture was gently mixed and incubated at 30 °C for 2 h. The resulting nanoparticles were cleaned twice with H_2O by spinning at $1000 \times g$ for 2 min and dispersed in 1 ml of H_2O .

Ensemble circular dichroism measurement in aqueous solution

The circular dichroism spectra were acquired using a JASCO J-1500 spectrophotometer with a 10 mm path length quartz at room temperature. A collection rate of 50 nm/min and a bandwidth of 1 nm were used during the measurement. SEM images were collected using a Hitachi S4800 SEM. The nanoparticles were drop-cast onto a silicon wafer and air-dried before taking SEM images.

Microscopic circular dichroism measurement in air

For the characterization of chiral gold nanoparticles, a super-continuum laser (SuperK Extreme, NKT Photonics) filtered by a tunable filter (SuperK VARIA, NKT Photonics) was used as the illumination source. The polarization of the illumination is controlled by the previously mentioned half-wave plate and quarter-wave plate. The reflection spectra of the chiral nanoparticles are captured by the spectrometer (LR1-B, ASEQ Instruments). The spectra were acquired 10 times with an integration time of 100 ms at 2-nm intervals using ten sampling boxes, and the resulting data were averaged. A reference spectrum from a bare glass substrate was also collected and subtracted as background.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Seongmin Im: Data curation (equal); Formal analysis (equal); Methodology (equal); Software (equal); Validation (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Wei Hong:** Data curation (equal); Formal analysis (supporting); Methodology (equal); Visualization (supporting); Writing – review & editing (equal). **Gayatri P. Chandran:** Data curation (supporting); Formal analysis (supporting); Methodology (supporting); Validation (supporting); Writing – review & editing (equal). **Xing Wang:** Funding acquisition (equal); Investigation (equal); Methodology (equal); Project administration (equal); Resources (equal); Supervision (equal); Writing – review & editing (equal). **Yang Zhao:** Conceptualization (equal); Funding acquisition (equal); Investigation (equal); Methodology (equal); Project administration (equal); Resources (equal); Supervision (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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